DRUGS · THYROID AND PARATHYROID

Radioactive iodine

Radioactive iodine (iodine-131) is taken up by thyroid tissue and destroys it with beta radiation. After total thyroidectomy for differentiated thyroid cancer, it ablates the remnant, treats microscopic or metastatic disease, and allows surveillance with thyroglobulin and scans. It also treats Graves' disease and toxic nodules. ENT surgeons meet its side effects, above all radioiodine sialadenitis, dry mouth and nasolacrimal obstruction.

1 CHAPTER ONE At a glance

The treatment on one page.

1.1 Summary

RADIOACTIVE IODINE

	SUMMARY
Agent	Iodine-131 (sodium iodide), given orally
How it works	Taken up through the sodium-iodide symporter; beta emission (range about 1 to 2 mm) destroys cells; gamma emission allows imaging
ENT uses	Remnant ablation after total thyroidectomy (selected low and intermediate risk patients); adjuvant treatment (intermediate and high risk); treatment of residual, nodal or distant metastatic disease; Graves' disease and toxic nodular goitre
Doses	Remnant ablation commonly 1.1 GBq (30 mCi) in low-risk disease; 3.7 GBq (100 mCi) or more for adjuvant or metastatic treatment; lower doses for benign disease
Preparation	Low-iodine diet for about 2 weeks; TSH stimulation (above 30 mU/l) by withdrawal of levothyroxine or recombinant TSH; avoid iodinated CT contrast in the preceding months; pregnancy test
Contraindications	Pregnancy and breastfeeding; inability to follow radiation protection precautions

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Who needs it after thyroidectomy

Decisions follow pathology and risk: not routinely in very low risk (unifocal tumours of 1 cm or less); considered in low to intermediate risk; recommended in high risk (gross extrathyroidal extension, distant metastases). Lobectomy alone is adequate for many low-risk cancers, and radioiodine requires a total thyroidectomy first.

2.2 Radioiodine refractory disease

Disease that does not take up iodine, or progresses despite it, is treated by surgery, external beam radiotherapy or targeted therapy (see targeted therapy for thyroid cancer).

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- **Radioiodine sialadenitis**: the salivary glands concentrate iodine; acute swelling and pain within days, and chronic sialadenitis with recurrent swelling, xerostomia and taste change in a proportion; risk rises with dose. Hydration and sialogogues around treatment may help; sialendoscopy can treat duct stenosis.
- **Nasolacrimal duct obstruction** with epiphora; dental caries from dry mouth.
- Nausea, neck swelling and pain (thyroiditis of remnant), transient bone marrow suppression, gonadal effects (avoid pregnancy for 6 months, conception for men for about 4 months), a small increase in second cancers with high cumulative doses, pulmonary fibrosis in diffuse lung metastases.

- In Graves' disease, radioiodine can worsen **thyroid eye disease**: steroid cover, especially in smokers.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Indications after thyroidectomy by risk.
- Preparation: low-iodine diet, TSH stimulation.
- **Sialadenitis** and other ENT side effects.

4.2 Common pitfalls

COMMON PITFALLS

- Giving iodinated contrast shortly before planned radioiodine.
- Missing radioiodine sialadenitis in a patient with recurrent parotid swelling.

Written for this app from BNF and BNF for Children; NICE and specialist society guidance; Scott-Brown; Cummings. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.